

Wiskott-Aldrich Syndrome

AT A GLANCE

- **Inheritance:** X-linked recessive
- **Gene:** Mutations in *WAS* (Wiskott-Aldrich syndrome gene)
- **Classic clinical triad:**
 - Recalcitrant dermatitis
 - Recurrent pyogenic infections
 - Hemorrhage due to thrombocytopenia and platelet dysfunction
- **Therapy:** Hematopoietic stem cell transplantation (bone marrow transplantation)

Immunological Features

Patients with Wiskott-Aldrich syndrome (WAS) exhibit multiple immune cell defects, including:

- Impaired **NK cell cytotoxicity**
- Defective **dendritic cell migration and activation**
- **Impaired macrophage chemotaxis**

These contribute to the broad susceptibility to infections and immune dysregulation seen in the disease.

Clinical Findings

The classic triad of hemorrhage, recurrent pyogenic infections, and recalcitrant dermatitis is present in only **25% of patients**. However, individual manifestations are common:

Bleeding Diathesis

- Present in **84% of patients**, making it the most frequent manifestation

- Typically appears in the **first weeks or months of life**
- Common presentations:
 - Bloody diarrhea (often initial sign)
 - Epistaxis
 - Hematemesis
 - Hematuria
 - Mucocutaneous petechiae
 - Intracranial hemorrhage

Thrombocytopenia is **persistent**, with platelet counts ranging from **1,000 to 80,000/ μ L**.

- **Diagnostic criterion:** Platelet count $< 70,000/\mu\text{L}$
- Platelets are characteristically **small (microthrombocytopenia)**
- Platelet aggregation is defective

Infections

- Recurrent **bacterial infections** begin in infancy as maternal antibodies wane
- Common infections:
 - Furunculosis
 - Conjunctivitis
 - Otitis media and externa
 - Pansinusitis
 - Pneumonia
 - Meningitis
 - Septicemia
- Predominant pathogens: **Encapsulated bacteria**
 - *Streptococcus pneumoniae*
 - *Haemophilus influenzae*
 - *Neisseria meningitidis*

- Increased susceptibility to:
 - Herpesviruses and other viruses
 - *Pneumocystis jirovecii* (formerly *P. carinii*)

Dermatitis

- Occurs in **~80% of patients**
- Typically develops in the **first few months of life**
- Often **severe and widespread**
- Commonly affects:
 - Face
 - Scalp
 - Flexural areas
- Features distinguishing it from typical atopic dermatitis:
 - More **exfoliative**
 - **Serosanguineous crusting**
 - Frequent **lichenification**
- Complications:
 - Secondary bacterial infection
 - **Eczema herpeticum**
 - **Molluscum contagiosum**
- Additional IgE-mediated conditions:
 - Urticaria
 - Food allergies
- Asthma

Autoimmune Manifestations

- Occur in **up to 40% of patients**
- Most common:
 - **Vasculitis** (20%) – affecting skin, GI tract, brain, heart

- **Autoimmune hemolytic anemia** (14%)
- **IgA nephropathy** (up to 10%)
- **Other immune-mediated conditions:**
 - Angioedema
 - Dermatomyositis
 - Pyoderma gangrenosum
 - Erythema nodosum
- **Hepatosplenomegaly is common**
- **Lymphadenopathy, transient arthritis, and joint effusions may occur**

Laboratory Findings

- **Thrombocytopenia with small platelets**
- **Immunoglobulin abnormalities:**
 - ↓ IgM (common)
 - ↓ IgG (sometimes)
 - Absent isohemagglutinins
 - ↑ IgA, IgE, IgD
- **Impaired humoral and cellular immunity contributes to infection susceptibility and autoimmunity**

Note: Hematopoietic stem cell transplantation is the only curative therapy and is recommended early in the disease course.